SURE-Tuned Covariance Shrinkage for Portfolio Selection

Senior Thesis Proposal

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Research Question

Can Stein's Unbiased Risk Estimate (SURE) be used to tune the shrinkage intensity in covariance matrix estimation for portfolio selection, and how does it compare to the standard Ledoit–Wolf formula — especially for shrinkage targets that have no closed-form solution?

Motivation

The sample covariance matrix is the standard input to portfolio optimization, but it behaves badly when the number of assets N is large relative to the number of observations T: it is near-singular and produces wildly unstable weights. DeMiguel, Garlappi and Uppal (2009) show that optimized portfolios built on it are routinely beaten out of sample by the naive $1/N$ rule.

The standard fix is to shrink toward a structured target, $\hat{\Sigma} = (1 - \rho)\mathbf{S} + \rho\mathbf{T}$. Ledoit and Wolf choose the intensity ρ by minimizing an *asymptotic* approximation to the risk. SURE (Stein, 1981) instead gives a *finite-sample unbiased* estimate of risk, computable from the data alone, which can be minimized numerically over ρ for any target. The question is whether that finite-sample property buys anything in practice — precisely in the small-T, large-N regime where the asymptotic approximation is weakest.

Plan

- **Derive** the SURE expression for linear shrinkage using the Stein–Haff identity for Wishart matrices, and check early whether it collapses to Ledoit–Wolf for a fixed

target.

- **Test three targets:** the identity and single-factor market model (both have Ledoit–Wolf benchmarks), and a sector-block-diagonal target, which has no closed-form constant and is the real test of whether SURE generalizes.
- **Benchmark against** the sample covariance, $1/N$, Ledoit–Wolf, and cross-validated ρ.
- **Simulate** across a grid of N and T (including $N > T$), using data-generating processes where each target is in turn the correct one.
- **Apply** to CRSP daily returns for S&P 500 constituents, rolling-window minimum-variance portfolios, with significance tested via Ledoit and Wolf (2011).

Reading List

- Stein, C. (1981). "Estimation of the Mean of a Multivariate Normal Distribution." *Annals of Statistics*, 9(6), 1135–1151.
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- Schäfer, J. & Strimmer, K. (2005). "A Shrinkage Approach to Large-Scale Covariance Matrix Estimation." *Statistical Applications in Genetics and Molecular Biology*, 4(1).
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